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Stronger-than-Lyapunov notions of matrix stability, or how “flowers” help solve problems in mathematical ecology

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Abstract

Persistent patterns of interactions in a multi-component system (e.g., intra- and inter-species relations in a community of n interacting species) may imply a number of formalizations as special, stronger-than-Lyapunov, notions of matrix stability, like D -stability, qualitative stability, Volterra–Lyapunov stability, and others. A variety of these notions, each having a certain motivation with regard to uncertainties inherent in model applications, constitute a hierarchical topology, sometimes very intricate and not yet well-understood, in a formal space of real $n \times n$ -matrices. As visible forms of this hierarchy, Matrix Flowers are suggested where ‘petals’ correspond to subsets of particular stability kinds, whose visible inclusion/intersection represent logical implication/junction. The Flowers are constructed under a few simple conventions, and, in the absence of ready characterizations, to draw a ‘petal’ often poses a challenging mathematical problem, whose solution may reveal a new biological knowledge of a general nature. Particular ‘petals’ concern the topics of strong and weak interactions in a community, ‘key’ species in its structure, diffusion instability in its spatial dynamics, where the flower plays a heuristic role in formulating a new problem or/and stimulating a new application.

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1. Introduction

It is well known that *Lyapunov* asymptotic stability in the linear approximation of a system of ODEs is equivalent to the property $\operatorname{Re} \lambda(A) < 0$ to hold for any eigenvalue $\lambda(A)$ of the Jacobian matrix $A = [a_{ij}]$, the matrix being also called *Lyapunov stable*, or just *stable*. Less known are many special notions of matrix stability motivated in applied areas, such as mathematical ecology, or more specifically, theoretical population dynamics in a biological community of n interacting species. The general idea behind any special motivation goes back to the fact the Jacobian matrix (often called *community matrix* in population contexts) also mirrors, by the very definition of its entries a_{ij} , the structure of interactions among the system components. Special notions like *sign* stability, *diagonal*, D -, aD -, and other kinds of stability are stronger than the Lyapunov one in the sense that a special notion requires matrix A to be *Lyapunov stable* simultaneously with a set of “analogous” matrices, the meaning of “analogous” being determined with regard to a particular kind of uncertainty, which always exists in the dynamical system. For example, sign stability, which had also entered the literature under the name of *qualitative* stability [69,54], means keeping the Lyapunov stability under any quantitative variation of the matrix entries which retains the qualitative (or *sign*) pattern of the matrix invariable; herewith “analogous” means equivalence of the qualitative patterns, leaving the magnitude of the entries uncertain.

Once we realize that any specific kind of matrix stability implies a particular set of “analogous”, specifically stable matrices in the space $\mathbf{R}^{n \times n}$ of real $n \times n$ -matrices, the particular set being a subset of $\mathbf{S}_n$, the set of Lyapunov stable matrices, the next logical step is to see how various particular subsets relate to each other. In particular, is there any logical implication between, or any intersection of, any two particular subsets? Traditional diagrams which are normally used in set theory to illustrate elementary operations over two sets can be augmented with two more conventions (or rules, see Section 6) in order that a diagram be adapted to matrices and turned to a “Flower”, with the “petals” corresponding to the particular stability subsets.

Most of the special stabilities were introduced, like the qualitative stability, about 40 years ago, during the systems analysis boom in science, and most of them are given with formal definitions which can hardly be verified for a given matrix because of the existence quantors (i.e., symbols $\exists$ and what they symbolize) or generality (over a continuum set) quantors (i.e., symbols $\forall$ and what they symbolize) inherent in the definitions. It means that a formal definition to a special kind of stability does

not represent a ready *characterization*, unlike, for instance, the *diagonal dominance*, a rare exclusion, which can be easily proved/disproved by checking n inequalities of the original definition:

$$a_{ii} > \sum_{j \neq i}^n |a_{ij}| \quad (1.1)$$

and which implies at least the Lyapunov stability of $-A$ by an eigenvalue localization criterion such as ‘Geršgorin disks’ (see, e.g., [24,25]). In many special cases, furthermore, an $n \times n$ -characterization can even hardly be (or has not yet been) developed from the definition, unlike the well-known Routh–Hurwitz criterion for the Lyapunov stability (see, e.g., [16]). The latter testifies whether all the leading principal minors are positive in the proper *Hurwitz matrix* which is composed, in a certain way, of (quite calculable) functions $E_i(A)$, the sums of the order i principal minors of the given matrix A .

A proper Flower can therefore assist in comprehension of the characterization problem for particular stabilities, and this was actually the way in which the first Flower was designed by the author [44] to comprehend the proper relation between *diagonal stability* and *diagonal quasi-dominance* in general real matrices and in the *normal* ones (Sections 7 and 8). Since publication of the 1993 book, some new extensions have been obtained and some new applications have been heuristically stimulated by the Flower. This paper presents a survey of mostly well-known and some recent results, following rather the logics of Flower construction (Section 6) than the chronology of the results. Thus, the 40-years-old theory of *sign-stable* matrices appears only in Section 9. Yet the heuristic role of Flower representations makes often the logical aspect coincident with the chronological one: further Sections 10–12 show how additional special notions of matrix stability were incorporated into a Flower and suggest further illustrations for the use of Flowers in defining new particular notions relevant to ecological applications.

Characterization problems and containment relationships were also surveyed by Hershkowitz [22] and Datta [13], which differ from (yet overlap with) the present survey in the collection of stability properties addressed. The latter focuses on generalizations deducible from the *inertia* concept, which is not considered here.

Matrix stabilities represent an abundant source of mathematical exercises, some of them being educational or/and important to comprehend the material. The corresponding statements in the text are marked below with the symbol $\square$.

2. Lotka–Volterra equations, feasible equilibria, and the community matrix

What is nowadays called a ‘Lotka–Volterra population equations’ represents a system of n nonlinear ODEs

$$dN_i/dt = N_i \left(\varepsilon_i - \sum_{j=1}^n \gamma_{ij} N_j \right), \quad i = 1, \dots, n, \quad (2.1)$$

defined on the non-negative orthant $\mathbf{R}_+^n$ of the Euclidean space $\mathbf{R}^n$ by means of the vector parameter $E = [\varepsilon_1, \dots, \varepsilon_n]^T$ and an $n \times n$ -matrix $-\Gamma = [-\gamma_{ij}]$ (the minus sign is introduced here to save the original notation of V. Volterra). System (2.1) is considered as a model for population dynamics in a community of n biological species, each having an intrinsic rate, ε_i , of natural increase and interacting with some other(s) in a way which is described by matrix $-\Gamma$. Element $-\gamma_{ij}$ of this matrix indicates the effect that the increment in N_j , the population size of species j , renders upon the growth rate, dN_i/dt , of species i ; when $i = j$, the element $-\gamma_{ii}$ indicates a self-regulation effect.

Since the times of Lotka [51] and Volterra [85], the Lotka–Volterra equations, in general, and the interaction matrix $-\Gamma$, in particular, have inspired many themes in theoretical ecology (see, e.g., [73]) laid down by the observation that already the pair of signs of symmetric entries $(\gamma_{ij}, \gamma_{ji})$ gives a basis to classify the types of pair-wise interactions between species i and species j in the community, well-known in ecology under such names as ‘mutualism’, ‘competition’, ‘predation’, and others (see, e.g., [64]). The fundamental idea, which was explicitly articulated much later [53,54,78] as the classics did not make use of the stability vocabulary, is that matrix $-\Gamma$ both reflects the structure of interactions among, and within each one of, n species in a community and provides for a mathematical tool to analyze stability in the model community dynamics, at least, in the linear approximation of system (2.1).

If, indeed, the system has a *feasible equilibrium*, i.e., a solution, $\mathbf{N}^* = [N_1^*, \dots, N_n^*]^T$, of the linear algebraic system

$$\Gamma \mathbf{N}^* = E \quad (2.2)$$

all of whose components are strictly positive ($N_i^* > 0$), then the Jacobian matrix A of system (2.1) linearized at $\mathbf{N}^*$ takes on the following form:

$$A \equiv [\partial(dN_i/dt)/\partial N_j|_{\mathbf{N}^*}] = \text{diag}\{\mathbf{N}^*\}(-\Gamma) \quad (2.3)$$

when the linearization is done in terms of deviations $N_i - N_i^*$ ($i = 1, \dots, n$), and it takes on

$$A_1 = (-\Gamma)\text{diag}\{\mathbf{N}^*\} \quad (2.3')$$

when the deviations are put dimensionless, i.e., $N_i/N_i^* - 1$ [78,44]. By diagonal similarity, matrices (2.3) and (2.3') have the same spectrum, and the well-known criterion for Lyapunov asymptotic stability in the linear approximation requires the spectrum to be located to the left of the imaginary axis: $\text{Re } \lambda(A) < 0$, or $A \in \mathbf{S}_n$, in our terms.

A stable feasible equilibrium is interpreted as stable coexistence of n species, and the dual role the *community matrix* plays in this “scenario”, i.e., describes the

structure and verifies the stability, makes it a key tool that opens the gate to many speculations in theoretical community ecology. In the cast of “actors”, however, there are at least two competitors for the key role, namely, the matrix $(-I)$ of interaction coefficients, as it arises immediately from the Lotka–Volterra equations, and the Jacobian matrix (2.3), as the mathematical meaning of its entries (the partial derivatives) gives exactly what should be meant under the ‘index of interaction’.

“Competitive coexistence” of different definitions has long been documented in the literature ([10,14,62] and references therein). It must have been mediated by relationship (2.3) between the two matrices, which provides apparently for the same classification of species intra- and inter-relations. But what should be considered the *community matrix* for other, non-Lotka–Volterra, population equations? And does (2.3) provide for the same results in stability analysis? While the answer to the former question is evident for a mathematician, i.e., the Jacobian of a linearized system [44], it is not readily accepted by theoretical ecologists [21]. The answer to the latter question is given by the notion of

3. D -stability

Let $D = \text{diag}\{d_1, \dots, d_n\}$ denote a *positive diagonal* $n \times n$ matrix, i.e., a matrix in which all entries on its main diagonal are positive ($d_i > 0$), while all the rest are zero, and let $\mathbf{D}_n$ denote the set of all such matrices (a multiplicative group of positive definite matrices $\square$).

Definition. Matrix A is called *D -stable* ($A \in \mathbf{DS}_n$) if DA is stable for any positive diagonal D , or, in more formal terms, if $DA \in \mathbf{S}_n \ \forall D \in \mathbf{D}_n$ ([2]¹; [69]). An equivalent definition considers multiplication with D from the right. $\square$

Any matrix $-D$ with $D \in \mathbf{D}_n$ is trivially D -stable. A less trivial example is any matrix $A_1 \in \mathbf{S}_2$ with the sign pattern

$$\text{Sign}(A_1) = \begin{bmatrix} 0 & - \\ + & - \end{bmatrix}, \quad (3.1)$$

which already provides for stability and remains the same in the product DA_1 .

Three immediate consequences of the definition are:

- (1) set $\mathbf{DS}_n$ is invariant w.r.t. multiplication by any $D \in \mathbf{D}_n$;
- (2) set $\mathbf{DS}_n$ is invariant w.r.t. matrix transposition (A^T) and inversion (A^{-1}) $\square$;
- (3) any D -stable matrix is Lyapunov stable, i.e., $\mathbf{DS}_n \subseteq \mathbf{S}_n$, since the identity matrix I is a particular case of D .

On the other hand, any stable matrix A_2 with the sign pattern

¹ With a stronger formal definition, which was modified later into the cited one above.

$$\text{Sign}(A_1) = \begin{bmatrix} + & - \\ + & - \end{bmatrix} \quad (3.2)$$

cannot be D -stable since a sufficiently large $d_1 \gg d_2$ can easily result in the product $\text{diag}\{d_1, d_2\}A_2$ having the trace positive, hence being not stable. D -stable 2×2 -matrices thus represent a proper subset of the stable ones, $\mathbf{DS}_2 \subset \mathbf{S}_2$, and this inclusion can be easily extended for any n [44] by means of matrices $A_2 \oplus (-I_{n-2})$, where $I_m = \text{diag}\{1, \dots, 1\}$, the identity matrix $m \times m$.

Although D -stability was originally motivated and studied in the field of mathematical economics [2,69], its motivation in mathematical ecology is apparent, first, in view of the aforesaid distinction between two definitions of the community matrix: due to equality (2.3), this is only within D -stable matrices that both matrix A and $(-\Gamma)$ are guaranteed to be stable simultaneously; otherwise one of them may be stable, but the other one may not. Second, D -stable matrices feature stability which cannot disappear under any variations in equilibrium population sizes (see (2.3)) while the interaction pattern $(-\Gamma)$ remains the same. In other words, once a feasible equilibrium exists in a system (2.1) with a D -stable matrix $(-\Gamma)$, it is locally stable. The mere stability of $(-\Gamma)$ alone is insufficient to guarantee this property. $\square$

To formulate a necessary condition of D -stability we need a class of matrices which can hardly be interpreted in a subject area, but which can be characterized directly from its

Definition. Matrix $A \in \mathbf{R}^{n \times n}$ is called a P_0^+ -matrix if all its principal minors are non-negative and there is a positive principal minor in each order k , $1 \leq k \leq n$ [29]. Notation $A \in P_0^+$ should be understood as $A \in P_0^+ \cap \mathbf{R}^{n \times n}$.

A more narrow class of P_0^+ , namely, P^+ - or just P -matrices, in which all the principal minors are positive, was defined, in an equivalent form, historically earlier under the name of ‘Hicks conditions’ [57], and this gave Quirk and Ruppert [69, p. 314] a reason to call the equivalent P_0^+ conditions ‘almost Hicksian’, in spite of existence of the more technical term [15].

To formulate a necessary condition for D -stability below, and in the sequel, we say that a matrix A belongs to a set $-\mathbf{X}$ iff the matrix $(-A) \in \mathbf{X}$.

Theorem 3.1. Any D -stable matrix is a $(-P_0^+)$ -matrix [69], or, in formal terms, $\mathbf{DS}_n \subset -P_0^+$.

The lack of D -stability in matrices (3.2) can therefore be explained as a violation of the necessary condition that $-a_{11} > 0$ in matrix $-A$, while the former reasoning bears actually the logic of how the necessity was proved. Note also that some authors deal with ‘positive D -stability’ in the sense that stability of A means all $\text{Re } \lambda(A) > 0$ [7,8], which would remove the minus sign from the above formulation; otherwise one had to consider the ‘signed principal minors’ in the definition of P_0^+ [11, p.

256]. In any case, different signs on the main diagonal already signify failure of a P_0^+ condition.

Non-sufficiency of the necessary condition given in Theorem 3.1 for D -stability is illustrated by a P_0^+ -matrix [36, p. 52]

$$A_3 = \begin{bmatrix} 1 & -5 & -1 \\ 2 & 2 & -1 \\ 9 & 0 & 0 \end{bmatrix}, \quad (3.3)$$

whose negative $-A_3$ is even not stable $\square$.

Until recently, characterizations (i.e., testable criteria) of D -stability were only known for $n \leq 3$ [7,11,44], for tridiagonal matrices [9] and for arbitrary acyclic matrices [5]. In general however, ‘characterizing the D -stable matrices’ remained ‘one of the prominent unsolved problems of matrix theory’ ([8, p. 237]; see also [13, p. 585]). The major reason is in the generality quantor ($\forall$) for matrix D in the definition: while low orders $n = 2, 3$ allowed one to reduce the Routh–Hurwitz criterion for matrix $D(-A)$ down to some testable conditions for any given matrix $-A$, the case $n = 4$ already posed an obstacle on this way.

It can be shown, indeed, that the formulation reduces along this way to the following

Lemma 3.1 (a D -stability criterion for real 4×4 -matrices [35, p. 1370]). *A matrix $(-A) \in \mathbf{R}^{4 \times 4}$ is D -stable if and only if $A \in P_0^+$ and all the following inequalities:*

$$E_1(DA) > 0, \quad (3.4)$$

$$E_1(DA)E_2(DA) > E_3(DA), \quad (3.5)$$

$$E_1(DA)E_2(DA)E_3(DA) > E_3(DA)^2 + E_1(DA)^2E_4(DA) \quad (3.6)$$

hold true for any $D \in \mathbf{D}_4^1 = \{D \in \mathbf{D}_4 : d_{11} = 1\}$.

Johnson [30, p. 13] supplied an equivalent formulation with a remark that it ‘is not numerically checkable’. We however present an approach whereby it can be really checked [35].

Note first that all $E_i(DA)$ are positive whence $A \in P_0^+$, hence inequality (3.5) follows actually from (3.6). Rearranging (3.6), we can see that Lemma 3.1 results in

Theorem 3.2 (a D -stability criterion for real 4×4 -matrices). *A matrix $(-A) \in \mathbf{R}^{4 \times 4}$ is D -stable if and only if $A \in P_0^+$ and*

$$f(x, y, z) > 0 \quad \text{for all positive } x, y, z, \quad (3.7)$$

where

$$\begin{aligned} f(x, y, z) &= E_1(DA)E_2(DA)E_3(DA) - E_3(DA)^2 - E_1(DA)^2E_4(DA), \\ D &= \text{diag}\{1, x, y, z\} \in \mathbf{D}_4^1. \end{aligned} \quad (3.8)$$

Thus, after verification of $A \in P_0^+$, checking the criterion means checking whether a certain real *polynomial* $f(x, y, z)$ of three variables is positive in (the interior of) the positive orthant $\mathbf{R}_+^3$. Note that the total degree of polynomial (3.8) is not higher than 6, while its degree w.r.t. each variable is not higher than 3. $\square$

Second, the number of variables in the polynomial at the above check is generally known, by the Tarski–Seidenberg theorem [81,74], to be successively reducible by variable-excluding algorithms in polynomial programming problems [60]. To check whether a given, finite-degree, real polynomial in one variable x is positive over the positive semi-axis represents a classical problem of algebra to be solved by elementary means. Therefore, third, let the following polynomial programming problem be considered with regard to the polynomial f defined in (3.8):

find the local extremes of z over the set

$$\Omega = \{f(x, y, z) = 0, x > 0, y > 0, z \in \mathbf{R}\{0\}\}. \quad (3.9)$$

Theorem 3.2 is proved to be equivalent to

Theorem 3.3 (a D -stability characterization for real 4×4 -matrices [35, p. 1370]).
A matrix $(-A) \in \mathbf{R}^{4 \times 4}$ is D -stable if and only if $A \in P_0^+$,

$$f(x, y, 1) > 0 \quad \text{for all positive } x, y, \quad (3.10)$$

where $f(x, y, z)$ is defined in (3.8), and

$$\text{none of the solutions to problem (3.9) is positive.} \quad (3.11)$$

(That is why the assumption $z > 0$ present in (3.7) is omitted in (3.9) as the statement would otherwise be tautological.)

Table 1
Examples of various titles for Volterra dissipative matrices

Titles (a matrix is):	Publication sources
Dissipative	[85,82,78,44]
Volterra–Lyapunov stable	[11,23]
Diagonally stable	[4,32,43,38,6]
Lyapunov diagonally stable	[38]
Positively D -dissipative	[68]
$\in A_1$	[71]
$\in S_w$	[80,79]

Theorem 3.3 gives a characterization indeed since:

- (i) condition (3.10) can be tested due to the low degree of $f(x, y, 1)$ w.r.t. x, y and a limited number of the ensuing elementary conditions to be checked;
- (ii) condition (3.11) can be tested via checking all the solutions to problem (3.9) to be obtained by an efficient algorithm of successive variable exclusion [60].

The above algorithm, which is, in essence, a modification of the Gauss method to polynomial equations, splits the problem, at each successive step, into a finite number of sub-problems in the lesser number of variables. At the final step, there appears a finite (yet great) collection of overdetermined systems of equations in one variable, the consistency of at least one of them serving a necessary and sufficient condition for a local extremum to exist in the original problem.

Ideologically, Theorem 3.3 could be generalized for any $n \geq 4$ [33,34], yet it can hardly serve as a characterization for $n > 4$ because of computational obstacles. In our experience [35,33], the number of sub-problems to be solved for $n = 4$ is $\sim 10^3$ and it appears to be computable, whereas $n = 5$ leads already to $\sim 10^{10}$ sub-problems, which can hardly be tackled in modern computers. In some cases, however, finer necessary conditions and finer sufficient conditions can still be checked which are obtained from those for a real polynomial in several variables to be positive in the positive orthant, the latter being produced by the method of quasi-homogeneous polynomial forms [61,37,33,34].

As regards simpler sufficient conditions, we shall see ‘the broadest known’ one [31, p. 89] to be just a definition of a stronger notion of stability, namely,

4. Diagonal stability, or Volterra dissipativeness

When studying systems of type (2.1), Volterra [85] distinguished two particular cases, namely, *conservative* and *dissipative* ones. Although his original definition concerned the systems of equations, it was actually formulated in terms of matrix Γ alone and, with the notation as above, it is equivalent to the following

Definition. Matrix $A \in \mathbf{R}^{n \times n}$ is called *Volterra dissipative* ($A \in \mathbf{DiS}_n$) if there exists a diagonal matrix $D \in \mathbf{D}_n$ such that matrix DA generates a negative definite quadratic form (DAx, x) , or equivalently, matrix $DA + A^T D$, the *symmetrizer* of DA , is negative definite.

Dissipative matrices are so popular among those studying stability in multi-component dynamical systems that many authors even proposed their original names for this matrix property (Table 1). A fundamental reason for the popularity is in the general Lyapunov theorem of matrix stability, which states the stability of matrix A to be equivalent to existence of a positive definite matrix H such that $AH + HA^*$

(with A^* denoting the Hermitian conjugate of A) be negative definite (see, e.g., [3, Chapter I.8]; [52, point III.3.3.7]). The dissipative matrices are thus a proper subset of stable ones for which matrix H can be just positive diagonal ($H \in \mathbf{D}_n$), and this serves a motivation for the title of ‘diagonal stability’.

In Lotka–Volterra population equations (2.1), if matrix $(-\Gamma)$ is dissipative and if a feasible equilibrium exists, then it is asymptotically stable globally in the positive orthant of $\mathbf{R}^n$ ([85]²; [78,44]). Moreover, on the positive orthant $\mathbf{R}_+^n$ with its boundary faces, there always exists a unique equilibrium which is globally stable in the interior of the orthant [80,23,79]. If one considers system (2.1) near the equilibrium point on a boundary face of $\mathbf{R}_+^n$, within the subspace $\mathbf{R}^m$ ($m < n$) of non-zero equilibrium components, then equations (2.1) reduce to the subsystem with a feasible equilibrium and the proper principal submatrix of $(-\Gamma)$. Any principal submatrix of a dissipative matrix is dissipative too $\square$, hence the boundary equilibrium is globally stable within $\mathbf{R}_+^m$. Therefore, once a feasible equilibrium exists in a system (2.1) with a dissipative matrix $(-\Gamma)$ or in any of its subsystems, it is globally stable.

Like D -stable matrices, the dissipative ones ($\mathbf{DiS}_n$) are also invariant w.r.t. multiplication by any $D \in \mathbf{D}_n$, matrix transposition (A^T) and inversion (A^{-1}) $\square$ [4,44]. As was noted above, $\mathbf{DiS}_n \subset \mathbf{S}_n$, which, in combination with the former invariance, results apparently in a stronger containment relation, namely, $\mathbf{DiS}_n \subseteq \mathbf{DS}_n$. If we remember now that neither sign-definite matrix can have a zero entry on its main diagonal, then D -stable matrices $A \oplus (-I_{n-2})$, with A_1 having the sign pattern (3.1), belong to $\mathbf{DS}_n \setminus \mathbf{DiS}_n$, thus indicating the proper inclusion

$$\mathbf{DiS}_n \subset \mathbf{DS}_n. \quad (4.1)$$

In particular, any principal submatrix of a dissipative matrix is D -stable too, which brings about a necessary condition of dissipativeness in the form of inclusion

$$(-\mathbf{DiS}_n) \subset P^+, \quad (4.2)$$

where P^+ is defined in the context of Theorem 3.1 of the previous section $\square$ [29]. Notation $A \in P^+$ should again be understood as $A \in P^+ \cap \mathbf{R}^{n \times n}$ (which is testable at least for fixed low n) and statement (4.2) as a logical contraction of Theorem 3.1.

As regards criteria, the explicit characterizations of dissipative matrices are only known, again, for $n \leq 3$ [11,38,44], though some authors did use the term ‘characterization’ when referring to some general results [38]. The characterization problem for a given numeric $n \times n$ -matrix in the case of general $n > 3$ was proved to be equivalent to 2 problems of order $(n-1)$ [71], thus being reducible to ‘an effective computational procedure’ [71, p. 612] of solving 2^{n-3} problems for matrices 3×3 .

Relationship (4.1) explains, on the one hand, why the definition of dissipativeness was considered for some time as ‘the strongest sufficient condition for D -stability’ [12, p. 135], yet the existence term of the definition itself can hardly be verified in a general case. On the other hand, (4.1) can be further refined with the notion of

² Although Volterra did not make use of stability terminology, his results on convergence of trajectories mean the global stability.

5. Total stability

Definition. Matrix A is called *totally stable* ($A \in \mathbf{TS}_n$) if any principal submatrix of A is D -stable [69]. An equivalent definition considers multiplication with D from the right. $\square$

An immediate consequence of the definition is that any principal submatrix of a totally stable matrix is totally stable too. When A is interpreted as a community matrix, its principal submatrix represents interactions in the subset of species that remains after some species are deleted from the community (under the hypothesis that deletion of a species does not affect the structure of interactions among (and within) the remaining ones). Therefore, total stability of a community matrix means conservation of this property after any group of species has been removed from the initial composition. This relates apparently to the so-called ‘species-deletion stability’ concept of theoretical ecology [66], although the latter implies just mere stability of the remaining species composition.

Related also is the theme of ‘community assembly sequences’: the final species composition depends crucially on a sequence in which potential colonizers do naturalize in the community [67,39]. Due to the globally stable pattern of dynamics in dissipative systems, mentioned in the previous section, the final outcome of colonization is always the same for the dissipative matrices. But the absence of any non-trivial outcome can be guaranteed by the totally stable matrices too, because each subset of species brings about a D -stable community submatrix, hence a feasible equilibrium which is (at least locally) stable irrespective of the equilibrium values.

This argument leads to a conjecture that

$$\mathbf{DiS}_n \subset \mathbf{TS}_n, \quad (5.1)$$

which would refine the containment relation (4.1). A proof that $\mathbf{DiS}_n \subseteq \mathbf{TS}_n$ follows from the inclusion (4.1) applicable for each submatrix. To prove the proper inclusion (5.1), it hereafter suffices to find a totally stable matrix which would not be dissipative.

When searching for such a matrix, $A_{\mathbf{TS} \setminus \mathbf{DiS}}$, one has to note that a similar refinement of (4.2) results in a (testable) necessary condition of total stability:

$$(-\mathbf{TS}_n) \subset P^+ \quad (5.2)$$

$\square$. In particular, neither totally stable matrix can have any zero entries on its main diagonal.

An apparent way to characterize a totally stable matrix is to check D -stability in a combinatorial number of its principal submatrices. Again, more explicit characterizations are known for $n \leq 3$ [42,44], and they reveal, for example, matrix

$$A_{\mathbf{TS} \setminus \mathbf{DiS}} = - \begin{bmatrix} 1 & 0.3 & 2 \\ 0.95 & 1 & 1.92 \\ 0.49 & 0.495 & 1 \end{bmatrix}^3 \quad (5.3)$$

to belong to $\mathbf{TS}_3 \setminus \mathbf{DiS}_3$. Hereby, $A_{\mathbf{TS} \setminus \mathbf{DiS}} \oplus (-I_{n-3}) \in \mathbf{TS}_n \setminus \mathbf{DiS}_n$.

³ Reproduced with a misprint in [44, p. 126].

To summarize, all the aforesaid containment relations constitute a true hierarchy:

$$\mathbf{DiS}_n \subset \mathbf{TS}_n \subset \mathbf{DS}_n \subset \mathbf{S}_n. \quad (5.4)$$

This hierarchy reduces just to equivalence when confined to *normal* matrices, i.e., to those which commute in the product with the transpose: $AA^T = A^T A$. To prove the equivalence, it suffices just to establish that $\mathbf{S}_n(N) \subseteq \mathbf{DiS}_n(N)$ for the normal matrices $\square$ [78,44]. The latter can be inferred from expressing the eigenvalues of $(A + A^T)/2$ via those of A based upon the fact that any normal matrix A can be reduced to the diagonal form by a unitary similarity transformation (see, e.g., [24, p. 101]).

To see the above hierarchy and equivalence in a visible form, we suggest what is called

6. Matrix Flower

The idea of Matrix Flower stems from the traditional Venn diagrams of set theory, in which a visible inclusion/intersection of graphic contours represents a logical implication/intersection among the corresponding sets of “points”. Each point represents a matrix $A \in \mathbf{R}^{n \times n}$ and each type of matrix stability is presented by a contour (*petal*) in the diagram. Two more conventions below distinguish a Flower from traditional Venn diagrams:

- (C1) There are three half-planes visible in the diagram (see Fig. 1), the upper vertical one containing matrices which have no zero entries on the main diagonal, the horizontal one which have all the diagonal entries non-positive and at least one zero. The lower vertical half-plane thus contains matrices which have at least one positive diagonal element.
- (C2) Those sets which are open in the natural topology of $\mathbf{R}^{n \times n}$ are pictured as curvilinear contours, those which are not open as rectangular contours.

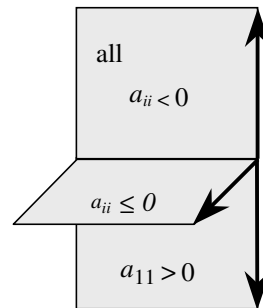


Fig. 1. Convention (C1).

Now, with a due account of all the aforesaid, the Flower diagram takes on the form shown in Fig. 2. Indeed, the basic pattern of the Flower is laid down by the hierarchy (5.4), which overlaps (and certainly agrees) with the containment diagram on p. 173 of Hershkowitz [22]. The set of stable matrices is apparently open due to continuity of polynomial roots as functions of the coefficients, hence (composite ones) of the matrix elements. The set of dissipative matrices is open too since the proper signs of the nested minors of $DA + A^T D$ are preserved within $U(A)$, a small neighborhood of A , thus certifying dissipativeness of any $A' \in U(A)$ with the same $D \in \mathbf{D}_n$. The dissipative petal has no horizontal part due to condition (4.2). The same is true (see condition (5.2)) for the set of totally stable matrices, which was proved to represent the topological interior of $\mathbf{DS}_n$ (Hartfiel [20] with an adjustment commented by Cain [8]), hence being a fortiori open.

The horizontal petal of D -stable matrices is not open because, in $\mathbf{R}$, any small vicinity of a zero entry on the main diagonal contains positive numbers, hence any small vicinity of such a matrix in $\mathbf{R}^{n \times n}$ contains matrices which cannot be D -stable. To prove non-openness of the vertical petal, one should consider a D -stable matrix which has a zero principal minor of order $m \geq 2$ $\square$.

The set of normal matrices is closed, yet the closure goes beyond the stable matrices $\square$. An (otherwise non-trivial) observation is that a normal stable matrix cannot have any zero on its main diagonal.

Finally, the two sets of matrices which have not yet been mentioned are

7. Quasi-dominant and M -matrices

Definition. Matrix A is called (row) *diagonally dominant* ($A \in \mathbf{DD}_n$) if the dominance condition

$$a_{ii} > \sum_{j \neq i}^n |a_{ij}| \quad (7.1)$$

holds in each row $i = 1, \dots, n$. Matrix A is called (row diagonally) *quasi-dominant* ($A \in \mathbf{qDD}_n$) if there exists a positive diagonal matrix $Q = \text{diag}\{q_1, \dots, q_n\}$ ($Q \in \mathbf{D}_n$) such that the product AQ is diagonally dominant ($AQ \in \mathbf{DD}_n$). Clearly, $\mathbf{DD}_n \subset \mathbf{qDD}_n$.

Diagonal dominance is a rare example of a notion whose definition provides immediately for characterization. It has long been popular both in theory [76,4] and applications [63,70,75]. In theory, it follows from ‘Geršgorin disks’ (see e.g., [24, p. 345]) that diagonal dominance in matrix $(-A)$ guarantees, at least, mere stability of A with the negative main diagonal. In fact, even quasi-dominance in A implies dissipativeness of $(-A)$ [58], so that $(-\mathbf{qDD}_n) \subseteq \mathbf{DiS}_n$, which is shown in Fig. 2.

Set $\mathbf{qDD}_n$ is evidently open due to the strict inequality in (7.1), but its characterization is far from evident because of the existence term in the definition. It however appears to go back to the characterization problem for the so-called

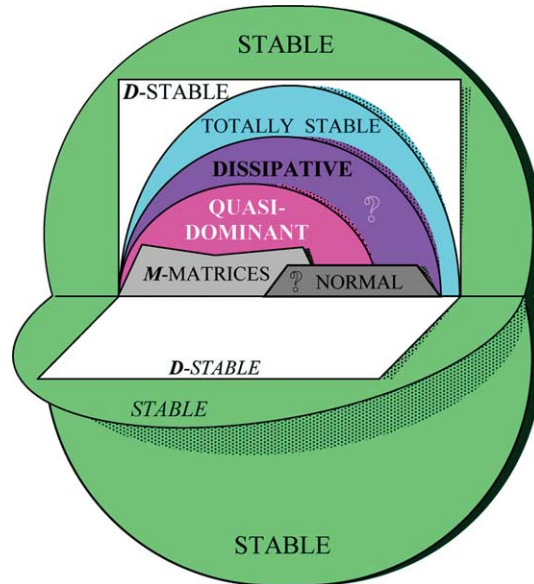


Fig. 2. Logical relations among matrix stabilities (adapted from [44, Fig. 16]).

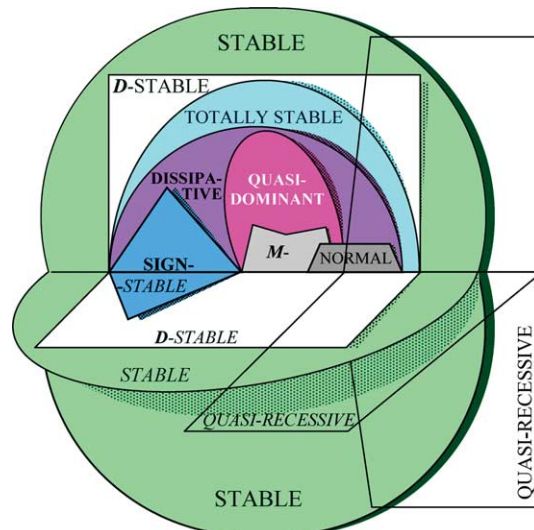


Fig. 4. Petals of sign-stable matrices within the Flower (adapted from [44, Fig. 30]).

M -matrices, motivated and extensively studied in the literature on mathematical economics [40,41]. Since being introduced by Ostrowski [65], the notion was defined in a number of equivalent ways, leading eventually to the definition that includes all of them [63].

Definition. Matrix $M = [m_{ij}] \in \mathbf{R}^{n \times n}$ with non-positive off-diagonal entries $m_{ij} \leq 0$ ($i \neq j$) is called a (non-singular) M -matrix ($M \in \mathbf{M}_n$) if any one of the following conditions holds:

- (M1) $\exists$ vector $\pi > \mathbf{0}$ such that $M\pi > \mathbf{0}$ (quasi-dominance of M);
- (M2) the linear system

$$\sum_{j \neq i}^n m_{ij} x_j = c_i, \quad i = 1, \dots, n, \quad (7.2)$$

is solvable in non-negative numbers x_j for any set of non-negative numbers c_j , $i = 1, \dots, n$;

- (M3) all nested (Sylvester) principal minors of M are positive;
- (M4) all principal minors of M are positive, i.e., $M \in P^+$.

Conditions (M3) and (M4) represent apparent characterizations, which also helps, due to their equivalence to (M1), in characterizing quasi-dominant matrices. This becomes clear from the observation that any given matrix $A = [a_{ij}]$ is (or is not) quasi-dominant simultaneously with its M -transformation $M(A) = [m_{ij}(A)]$, where

$$m_{ij}(A) = \begin{cases} a_{ii}, & i = j, \\ -|a_{ij}|, & i \neq j. \end{cases} \quad (7.3)$$

It follows hereby that

$$\mathbf{M}_n \subset \mathbf{qDD}_n, \quad (7.4)$$

as is also shown in Fig. 2. When confined to M -matrices, all the stabilities of larger nested petals become equivalent. The equivalence diagram of Hershkowitz [22, p. 174] is thus expressed in the $\mathbf{M}_n$ petal being visibly included into the others. Set $\mathbf{M}_n$ is not open since the fixed sign pattern of an M -matrix is disturbed in any small vicinity of any zero element. Normal non-singular M -matrices may also be found $\square$, thus giving a positive answer to the question marked between the two petals.

Whether or not the proper inclusion $(-\mathbf{qDD}_n) \subset \mathbf{DiS}_n$ holds true was not clear for some time, which is designated with another question mark between the two petals.

In applications to population dynamics, diagonal dominance in the (negation of) the community matrix is interpreted as species self-regulation effects being stronger than interactions among species [55,44]. In the absence of an explicit answer to the question posed in the Flower, the stronger self-regulation has been long perceived as a criterion, rather than just a sufficient condition, for global stability in a model

community [17,18,86,1]. This delusion persisted until the positive answer to the above question, reformulated as ‘Do there exist diagonally stable matrices without dominating diagonal?’ was obtained [43] by means of

8. Quasi-recessive matrices

To find a matrix of interest, we define first a property which is essentially opposite to the diagonal dominance.

Definitions. Matrix $A = [a_{ij}]$ is called *diagonally recessive in rows* if $\forall$ row $i \exists$ an off-diagonal entry a_{ik} such that $|a_{ii}| \leq |a_{ik}|, k \neq i$.

Matrix $A = [a_{ij}]$ is called *diagonally recessive in columns* if $\forall$ column $j \exists$ an off-diagonal entry a_{kj} such that $|a_{jj}| \leq |a_{kj}|, k \neq j$.

Matrix A is called *recessive* if it is diagonally recessive in rows or in columns.

Matrix A is called *quasi-recessive* ($A \in \mathbf{qR}_n$) if $\exists Q = \text{diag}\{q, \dots, q_n\} \in \mathbf{D}_n$ such that AQ is recessive.

Theorem 8.1 [43].

$$\mathbf{qR}_n \cap \mathbf{qDD}_n = \emptyset. \quad (8.1)$$

It means that the definition of quasi-recession is correct in essence, yet there may well be matrices which are neither quasi-recessive, nor quasi-dominant $\square$ [44]. Set $\mathbf{qR}_n$ is wide enough to contain even non-stable matrices and it is not open (closed actually) in the natural topology of $\mathbf{R}^{n \times n}$ $\square$.

In order to see whether there are dissipative matrices outside the quasi-dominant set, we may look for such matrices among those which are quasi-recessive, i.e., within the intersection of these two sets, or “petals”. In other, metaphoric words, we have to see what is the cross-section of the Flower by the “knife” of quasi-recessive matrices. Note first that Theorem 8.1 forbids the “knife” to reach the quasi-dominant petal. Whether it reaches the dissipative petal, or whether

$$\mathbf{qR}_n \cap \mathbf{DiS}_n \neq \emptyset \quad (8.2)$$

is another formulation of the above question. The positive answer is given, for example, with a matrix [44, p. 132]

$$A_{3R} = - \begin{bmatrix} 1 & 7/6 & 1/2 \\ 3/4 & 1 & 3/2 \\ 3/2 & 1/2 & 1 \end{bmatrix}, \quad (8.3)$$

which is apparently recessive (both in rows and columns) and which enters $\mathbf{qR}_3$ together with a neighborhood $U_1(A_{3R}) \subset \mathbf{qR}_3$. A known 3×3 -characterisation verifies A_{3R} to be dissipative [44], and it can be certified directly by definition with $D_3 =$

$\text{diag}\{5, 4, 40/9\}$. Again, there is a neighborhood $U_2(A_{3R}) \subset \mathbf{DiS}_3$, hence $U_1(A_{3R}) \cap U_2(A_{3R}) \subset \mathbf{qR}_3 \cap \mathbf{DiS}_3$, which illustrates statement (8.2) for $n = 3$. The expansion for $n > 3$, with matrix $A_{3R} \oplus (-I_{n-3})$, is no longer valid as $I_{n-3} \notin \mathbf{qR}_{n-3}$, but it rather illustrates the part of $\mathbf{DiS}_n$, outside $\mathbf{qDD}_n$ and $\mathbf{qR}_n$. It remains a matter of exercise to prove (8.2) in a general case $\square$.

Thus, there is really enough space between the dissipative and quasi-dominant petals of the Flower, and Fig. 3 shows the cross-section of interest (assuming the “knife” to be “transparent”). It suggests that $\mathbf{qR}_n \cap \mathbf{DiS}_n$ contains some normal matrices too $\square$, while there is also a small petal (implicit in Fig. 3) between the $\mathbf{qDD}_n$ and $\mathbf{qR}_n$ ones, within the trapezoid of normal matrices $\square$ [44]. Simpler exercises should confirm certain parts in common with $\mathbf{qR}_n$ and other petals shown in the Flower $\square$.

So, the Flower demonstrates vividly that the set of dissipative matrices is far from exhausted by quasi-dominant ones. Found with a heuristic aid of the quasi-recessive set, matrix (8.3) is interpreted as the community matrix for a competition community with certain relationships among intra- and inter-species competition coefficients. As a dissipative matrix, it witnesses—together with quite non-empty set $\mathbf{DiS}_n \setminus \mathbf{qDD}_n$ —that global stabilization of the community dynamics may not be necessarily achieved with stronger self-regulations: stronger interactions among species, as well as an appropriate combination of the two categories in the community structure, may also provide for the global stability.

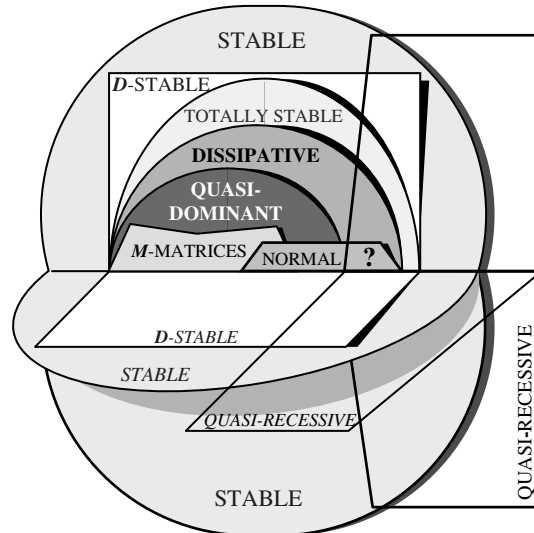


Fig. 3. Some quasi-recessive matrices are there between the dissipative and quasi-dominant ones (adapted from [44, Fig. 17].

Of practical interest is therefore a general characterization of quasi-recessive matrices. A fruitful approach to the problem makes use of the correspondence between paths and cycles in the associated digraph and those in the matrix [56].

Definition. A product of elements of matrix $A = [a_{ij}]$ is called *path* if the elements constitute a directed path $\vec{p} : i_1 \rightarrow i_2 \rightarrow \cdots \rightarrow i_l$ in the digraph, $\mathcal{D}(A)$, associated to A , i.e.,

$$A[\vec{p}] = a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_{l-1} i_l}, \quad (8.4)$$

and a *cycle* of matrix A if the elements constitute a cycle $\vec{c} : i_1 \rightarrow i_2 \rightarrow \cdots \rightarrow i_l \rightarrow i_1$ in $\mathcal{D}(A)$, i.e.

$$A[\vec{c}] = a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_{l-1} i_l} a_{i_l i_1}. \quad (8.5)$$

If the integers $i_1, i_2, \dots, i_l$ are all different, or if the correspondent path (cycle) has no self-intersections in the digraph $\mathcal{D}(A)$, then the path (cycle) of matrix A is called *simple*.

Theorem 8.2 [48]. *Non-negative irreducible matrix A with entries $\delta_i = \{0, 1\}$ ($i = 1, \dots, n$) on its main diagonal is quasi-recessive if and only if the condition*

$$A[\vec{c}] \geq \prod_{i \in V[\vec{c}]} \delta_i \quad (8.6)$$

holds true at least for one simple cycle $\vec{c}$ in A (here $V[\vec{c}]$ is the subset of vertices in $\mathcal{D}(A)$ which are spanned by the cycle $\vec{c}$).

Condition (8.6) is simple to check: its right-hand side equals 1 or 0; since the left-hand side is always positive and since, in a strongly connected digraph $\mathcal{D}(A)$, any vertex belongs to at least one cycle [19], Theorem 8.2 yields a simple sufficient condition as

Observation 8.1 [48]. *If an irreducible matrix has a zero entry on its main diagonal, it is quasi-recessive.*

The non-negative constraint of Theorem 8.2 does apparently not restrict its application as any matrix $[a_{ij}]$ enters $\mathbf{qR}_n$ simultaneously with $[|a_{ij}|]$. Nor does the unitary constraint on the diagonal entries as $\mathbf{qR}_n$ is invariant w.r.t. multiplication with any $D \in \mathbf{D}_n$. For a reducible matrix to be quasi-recessive, it is sufficient that so is its every irreducible diagonal block $\square$.

Other subsets of stable matrices can also be incorporated to the Flower, in particular,

9. Sign-stable matrices

Following the comments given in the Introduction, we define real matrices A and B to be *sign-equivalent* if they have the same sign pattern: $\text{Sign}(A) = \text{Sign}(B)$. Sign equivalence is a standard equivalence relation $\square$, so that $\mathbf{R}^{n \times n}$ splits up to (a finite number of) equivalent classes.

Each class can be considered as a certain coordinate orthant in a subspace of non-zero matrix elements, with either positive or negative directions of the corresponding axes.

Definition. Matrix A is called *sign-stable* ($A \in \mathbf{SS}_n$) if any matrix B sign-equivalent to A is stable [69,26].

It suggests that there are orthants in $\mathbf{R}^{n \times n}$ whose interiors contain only stable matrices, while a characterization of sign-stable matrices is supposed to indicate such orthants.

Under the name of ‘qualitative stability’ the notion was motivated in and introduced to mathematical economics [72,69], then reformulated in ecological terms [54] as a kind of stability which is determined exclusively by the qualitative pattern of (pair-wise) interactions in the community, irrespective of any quantitative uncertainties inherent in the interaction coefficients. After Quirk and Ruppert [69], the conditions they had published as a verifiable criterion of sign stability were launched into ecological literature [54] and were soon criticized scrupulously by Jeffries [26]. He demonstrated the conditions to be only necessary but not sufficient ones for sign stability of any irreducible matrix A and proved an additional condition, in terms of vertex colorings for $\mathcal{D}(A)$, which complemented the former conditions to being sufficient.

The complemented set of conditions was later proved to be also necessary [27, 28,49], which closed the characterization problem and provided for complete description of sign-stable patterns in ecological terms [50,44]: these are prey-predator structures with special allocation of self-limited species (i.e., negative entries on the main diagonal) or any collections of such structures linked by unilateral links forming no cycles in the total community digraph. The latter circumstance restricted drastically direct applications of the sign stability criterion in ecology, a science of turnovers, yet the criterion promoted a progress in theoretical speculations [44]. In particular, the color test indicated a key role that allocation of self-regulating species in the structure of interactions can play in its qualitative stability.

To delineate the petal(s) of sign-stable matrices in the Flower, note first that set $\mathbf{SS}_n$ is neither closed $\square$ (consider a sequence of sign-stable matrices with negative diagonal entries approaching zero), nor open $\square$ (consider a vicinity of zero elements whose non-zero values would form a cycle in the digraph). It contains both matrices with all negative diagonal entries (e.g., $-I_n$) and those with some zero diagonal

entries (e.g., matrices (3.1)). In the both cases, definitions reveal that $\mathbf{SS}_n \subset \mathbf{DS}_n$, while, in the former case, a finer inclusion is given by

Theorem 9.1 [42,44]. *If $A = [a_{ij}] \in \mathbf{SS}_n$ and $a_{ii} < 0 \forall i = 1, \dots, n$, then $A \in \mathbf{DiS}_n$.*

Therefore, the Flower with $\mathbf{SS}_n$ should look like shown in Fig. 4.

Among those special notions of matrix stability which are not shown in the Flower there is

10. Strong stability, or aD -stability

Definition. Matrix A is called *additively D -stable* [32] or *aD -stable* ($A \in \mathbf{aDS}_n$) if $A - D$ is stable for any non-negative diagonal D , or, in more formal terms, if $A - D \in \mathbf{S}_n \forall D \in \mathbf{D}_n^0$, where $\mathbf{D}_n^0 = \{D = \text{diag}\{d_1, \dots, d_n\} \mid d_i \geq 0 \forall i = 1, \dots, n\}$.

Under the name of *strong stability* the notion was suggested by Cross [11, p. 253], who referred just to ‘the theory of differential equations’ as an origin. Meanwhile, the motivation of the term might be the following. In the so-called ‘diffusion’ model of population dynamics [78, Section VIII.6] or ‘reaction–diffusion’ equations (see, e.g., [59])

$$\partial N_i / \partial t = d_i \nabla^2 N_i(\mathbf{x}, t) + f_i(\mathbf{x}, t); \quad d_i \geq 0, \quad i = 1, \dots, n, \quad (10.1)$$

where $N_i(\mathbf{x}, t)$ is the population or physical density, respectively, of species or substance i at a spatial point $\mathbf{x}$ and $d_i \geq 0$ is the (species- or substance-specific) rate of diffusion, a spatially homogeneous solution $N^*(\mathbf{x}, t) \equiv N^* > 0$ exists apparently if the feasible equilibrium N^* does so in the local or ‘reaction’ model

$$dN_i/dt = f_i(\mathbf{x}, t), \quad i = 1, \dots, n. \quad (10.2)$$

Mere stability of the Jacobian matrix A of (10.2) at N^* can no longer guarantee stability of the $N^*(\mathbf{x}, t) \equiv N^*$ w.r.t. any spatially periodic perturbation

$$n_i(\mathbf{x}, t) = n_i^0 \cos((\mathbf{k}, \mathbf{x}) + \varphi) \exp(\lambda t) \quad (i = 1, \dots, n), \quad (10.3)$$

with a wave number $k^2 = |\mathbf{k}|^2 \geq 0$. The latter kind of stability is only provided if the matrix

$$[a_{ij} - d_i k^2 \delta_{ij}] = A - D \quad (i, j = 1, \dots, n) \quad (10.4)$$

is stable for any non-negative values of $d_i k^2$ i.e., for any diagonal $D \in \mathbf{D}_n^0$. $\square$

Thus, stability in a strongly stable local model is ‘strong’ enough to resist the spatial perturbations. Meanwhile, it is the loss of stability by the homogeneous spatial distribution which was suggested as a general mechanism for non-trivial, spatially inhomogeneous, structures to emerge in morphogenesis [84,59] even in a homogeneous medium. Thereby, strongly stable matrices specify situations where the mechanism cannot work. However, since $\mathbf{aDS}_n \supseteq \mathbf{DS}_n$, at least for $n \leq 3$ [11], ‘strong’

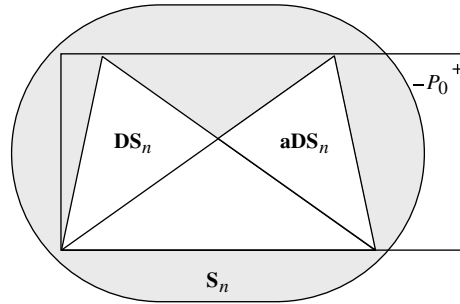


Fig. 5. Relation between D - and aD -stability for $n \geq 4$. Intersection of two triangles gives rise to the notion of DaD -stability.

stability is perhaps the weakest concept among all the special notions of matrix stability (Fig. 4).

A widely-known necessary condition for aD -stability is the same as that one for D -stability [11,36]:

$$\mathbf{aDS}_n \subset -P_0^+, \quad (10.5)$$

as is the 2×2 characterization:

$$\mathbf{aDS}_2 = \mathbf{DS}_2 = -P_0^+, \quad (10.6)$$

while the 3×3 characterization is somewhat simpler than that for D -stability:

$$\mathbf{aDS}_3 = \mathbf{S}_3 \cap (-P_0^+) \supset \mathbf{DS}_3. \quad (10.7)$$

This containment is no longer valid for $n \geq 4$: although both

$$\mathbf{aDS}_n \subset \mathbf{S}_n \cap (-P_0^+) \quad \text{and} \quad \mathbf{DS}_n \subset \mathbf{S}_n \cap (-P_0^+), \quad (10.8)$$

Togawa [83] found a D -stable 4×4 -matrix, A_{Tgw} , which is not aD -stable. Thereafter, matrix $A_{\text{Tgw}} \oplus (-I_{n-4})$ extrapolates the situation to greater n , and the corresponding fragment of the Flower should look as shown in Fig. 5 ($\mathbf{aDS}_n$ is neither open, nor closed $\square$).

This picture prompts a new petal to emerge as $\mathbf{DS}_n \cap \mathbf{aDS}_n$ and a respectively new notion of

11. DaD -stability

Definition [47]. Matrix A is called DaD -stable ($A \in \mathbf{DaD}_n$) if it is both D - and aD -stable (i.e., $\mathbf{DaD}_n = \mathbf{DS}_n \cap \mathbf{aDS}_n$).

Set $\mathbf{DaD}_n$ is invariant w.r.t. both multiplication with any $D \in \mathbf{D}_n$ and subtraction of any $D \in \mathbf{D}_n^0$. Its characterization represents a trivial consequence of those for D - and aD -stability for $n \leq 3$ and a less trivial combination of algorithms to characterize these subsets for $n \geq 4$.

DaD-stability has appeared to make sense in application to modelling global biogeochemical cycles [46], or in a more general context of compartment modelling [87], as a property of ‘Svicobians’. A *Svicobian* is a term that was proposed [46,47] for the Jacobian matrix in a wider-than-linear class of dynamic models behind a given ‘stores–flows’ diagram [77]. It takes on the following invariant form:

$$A = (S - Q)D, \quad (11.1)$$

where matrix $S = F^T - F$ is skew-symmetric by construction, with $F = [f_{ij}]$ and f_{ij} meaning the flow (of a substance) from compartment i to j ; $Q = \text{diag}\{q_1, \dots, q_n\}$ is a matrix of (constant) inputs $q_i \geq 0$ from outside, and $D = \text{diag}\{1/x_1^*, \dots, 1/x_n^*\}$ is a matrix of inverse equilibrium values for the stores $x_i^* > 0$, $i = 1, \dots, n$. If a Svicobian appears to be *DaD*-stable, it means that stability is independent of quantitative estimates of inputs and stores but is determined by the pattern of the flow diagram alone.

For a real $n \times n$ -matrix to be representable as a Svicobian (10.1), a characterization is obtained in terms of simple cycles in the matrix [48]. For a Svicobian to be *DaD*-stable, failure of the Jeffries’ color test [26] is proved to be sufficient [47], and examples of stable Svicobians are found which are not *DaD*-stable. Since the color test is of pure qualitative nature, its failure appears to be sufficient for equilibrium stability in the model to be independent of any quantitative characteristics. This serves additional heuristics in selection of models to be used, for instance, in choosing a better version of aggregating a given ‘stores–flows’ diagram up to greater compartments [46].

Additional heuristics relate also to the notion of

12. Structurally open subsets

Because of uncertainties or/and errors inherent in practical estimation of matrix elements, the practical meaning of openness is evident. However, the classical definition of an open set considers a vicinity of the point to be an open set in the basic topology of the space. In case of matrices it means that any matrix element, including zero ones, may be perturbed. While perturbations of non-zero elements are logically interpreted as data errors in experiments or observations, those of zero elements should mean introduction of the correspondingly new links into the system, i.e., certain changes in the system structure. If however the structure is assumed fixed in the study, then only non-zero elements can be perturbed in observation/experiments. Any fixed subset of matrix elements constitute a corresponding subspace in $\mathbf{R}^{n \times n}$, and this brings about the following

Definition [45]. A subset $\mathbf{X} \subset \mathbf{R}^{n \times n}$ is called *S-open (structurally open)* if any point, i.e., a real $n \times n$ -matrix $A \in \mathbf{X}$, is interior in the relative topology of the subspace, $\mathbf{R}_\#(A)$, of all the non-zero elements in A . In more formal terms, if $U_\#(A)$ denotes a vicinity of A in the relative topology of $\mathbf{R}_\#(A)$, then a subset $\mathbf{X}$ is called *S-open* if $\forall A \in \mathbf{X} \exists U_\#(A) \subset \mathbf{X}$.

Two immediate consequences of the definition are:

Corollary 12.1. *Any subset that is open in $\mathbf{R}^{n \times n}$ is also S -open.*

Yet there may well be subsets which are S -open but not open in $\mathbf{R}^{n \times n}$, the subset of sign-stable matrices serving an immediate example, where perturbations of the non-zero elements may even not be small. $\square$

On the other hand, the non-open set of D -stable matrices is proved to be not S -open either [45], and similar reasoning can prove the aD -stable matrices to be not S -open, too.

If now the classical notion in the Flower convention (C2) is changed into more practical one of open subsets, then only the sign-stable petals transform into curvilinear contours $\square$.

13. Conclusion

The variety of stronger-than-Lyapunov notions of matrix stability considered in this survey is certainly not exhaustive. We have rather illustrated how such a notion can originate from a subject area, mostly due to uncertainties inherent in model applications, and pose mathematical problems in matrix theory and similar disciplines. Some of these problems have been solved, others await solution. The logical hierarchy of special stability notions is established in a formal space of real $n \times n$ -matrices as the Matrix Flower, a visible form of traditional Venn diagrams adapted to stable matrices under a few simple conventions. In the absence of immediate explicit characterizations, to draw a ‘petal’ in the Flower means to solve a mathematical problem, whose interpretation in ecological context may reveal new knowledge of a general nature. As a general picture with specific features, the Flower is of didactic value in comprehension of stability concepts and it plays a heuristic role in formulating new problems or/and inducing new applications.

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